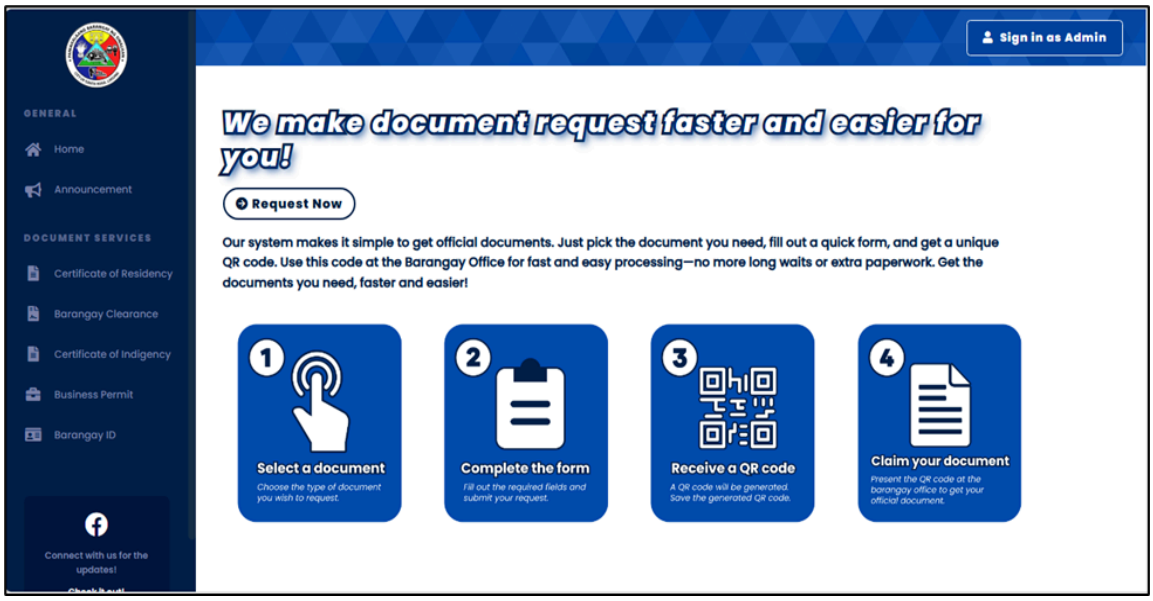
Prototype

This section presented the prototype design and functionalities of the Web-Based Document Issuance System with QR Code Technology. It outlined the key features of the system, including its user interface interactions and the underlying processes across various sections. This section provided a comprehensive overview of how the system's components integrate effectively to optimize document issuance, ensure efficient record-keeping, and enhance user management through the use of QR code technology.

Figure 1. Homepage



The Homepage offers a brief overview of the system's features and includes a step-by-step guide to help users understand how it works. It features an Announcement Section for important updates and a Document Services Section where users can view the available services. The "Request Now" button provides easy access by redirecting users directly to the Document Services

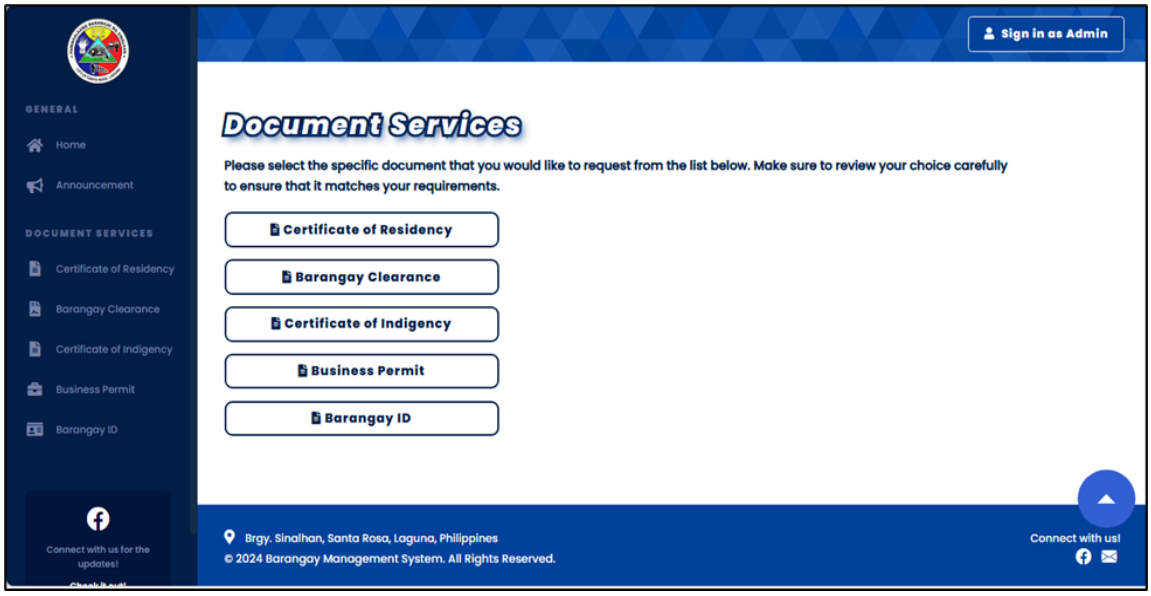
Section. The fixed header includes a “Sign in as Admin” button, which takes users to the login page specifically for admin or staff members, ensuring secure access to administrative functions. Additionally, a fixed sidebar with quick navigation links ensures the interface remains consistent and user-friendly throughout all pages. Each resident page also includes a footer that contains the contact details of the barangay, providing easy access to communication for residents in need of assistance.

Figure 2. Announcement Section



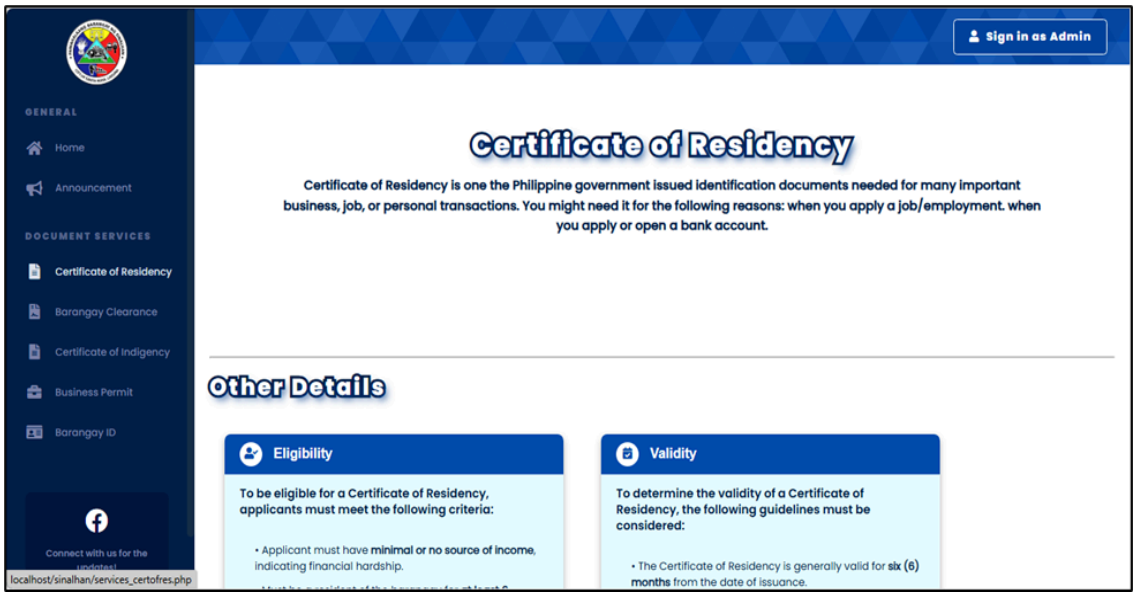
The Announcement Section, located on the Homepage, displays announcement messages posted by the admin or staff. Each announcement includes details such as the poster's name, the date, and the time it was posted. Announcements will automatically disappear once they are removed by the admin or staff.

Figure 3. Document Services Section



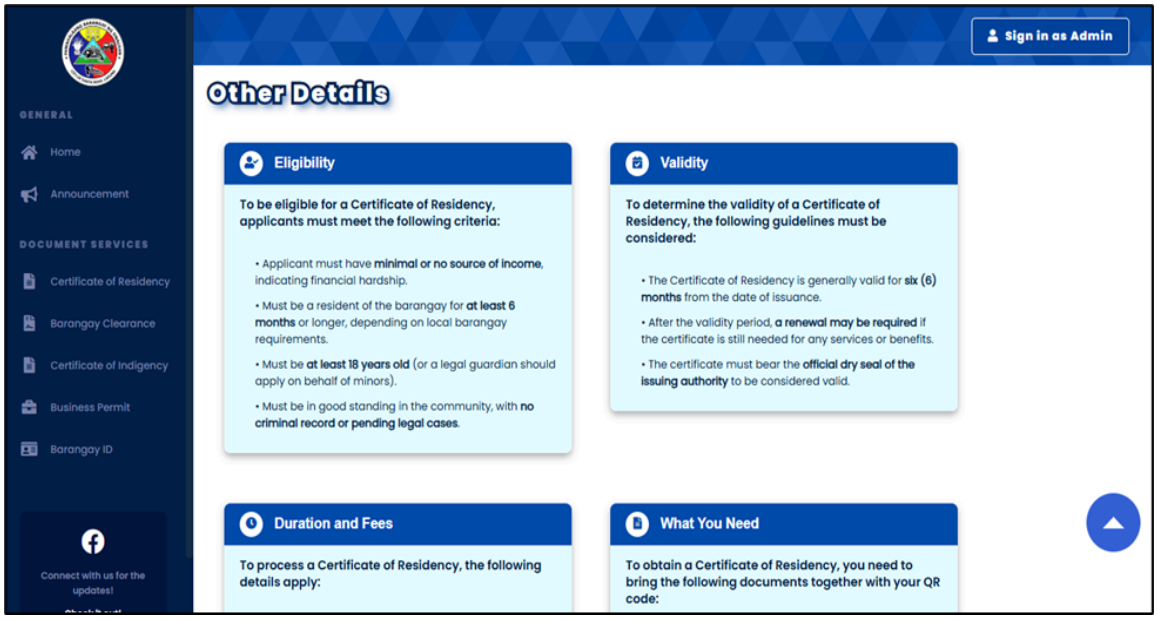
The Document Services Section contains the various document services offered by the barangay. It consists of buttons labeled with the type of document available, making it easy for residents to navigate. When a button is clicked, the system redirects the resident to the specific webpage for the selected document.

Figure 4. Certificate of Residency Request Page



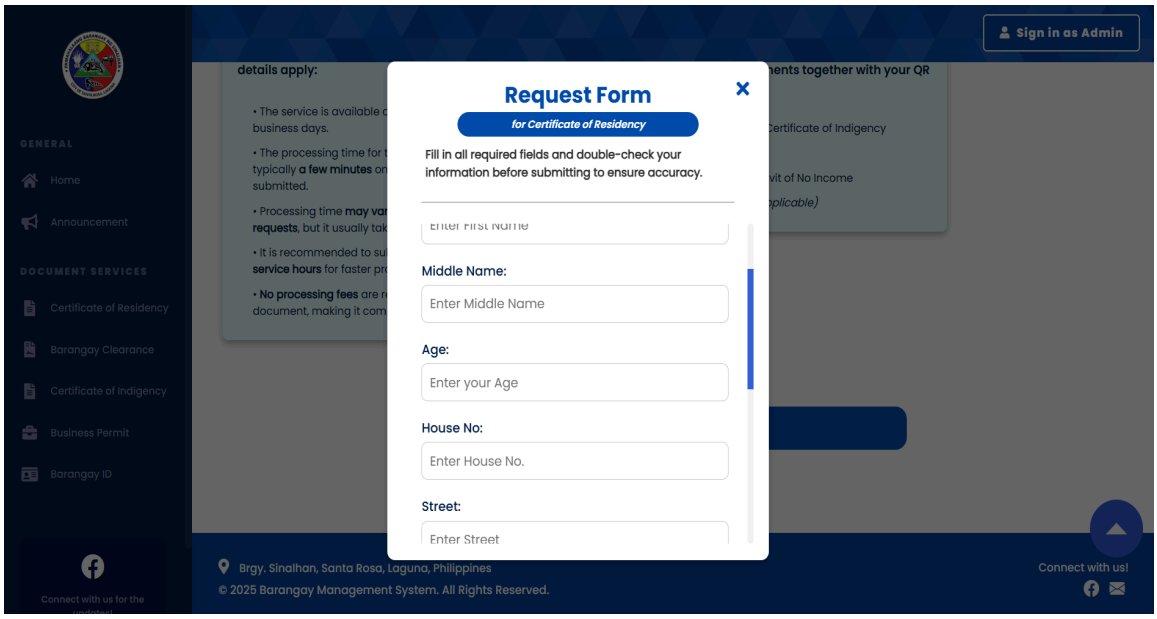
Each document type available for request has its own dedicated page. This page provides essential information, including a brief description of the document and other relevant details. It is also where residents can fill out a request form to generate a QR code for their document request.

Figure 5. Other Details Section



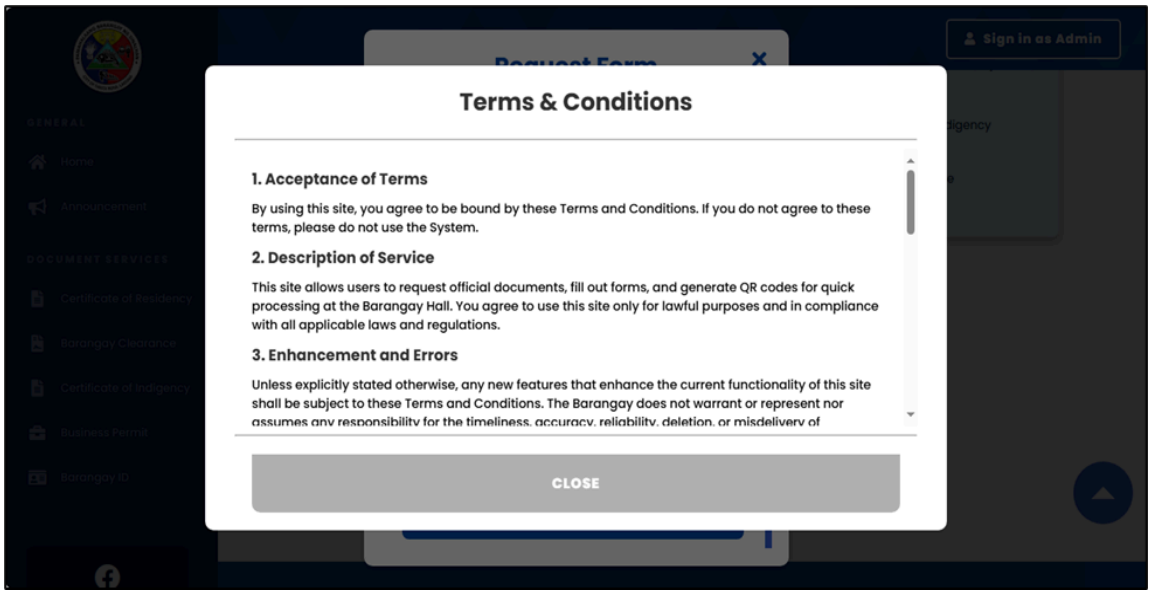
The Other Details Section for each document type provides information about its validity, eligibility criteria, processing time, and applicable fees. It also includes a list of requirements that residents need to bring, along with the generated QR code, when claiming their requested document at the barangay hall. Residents must review this section before proceeding with the request.

Figure 6. Request Form Popup



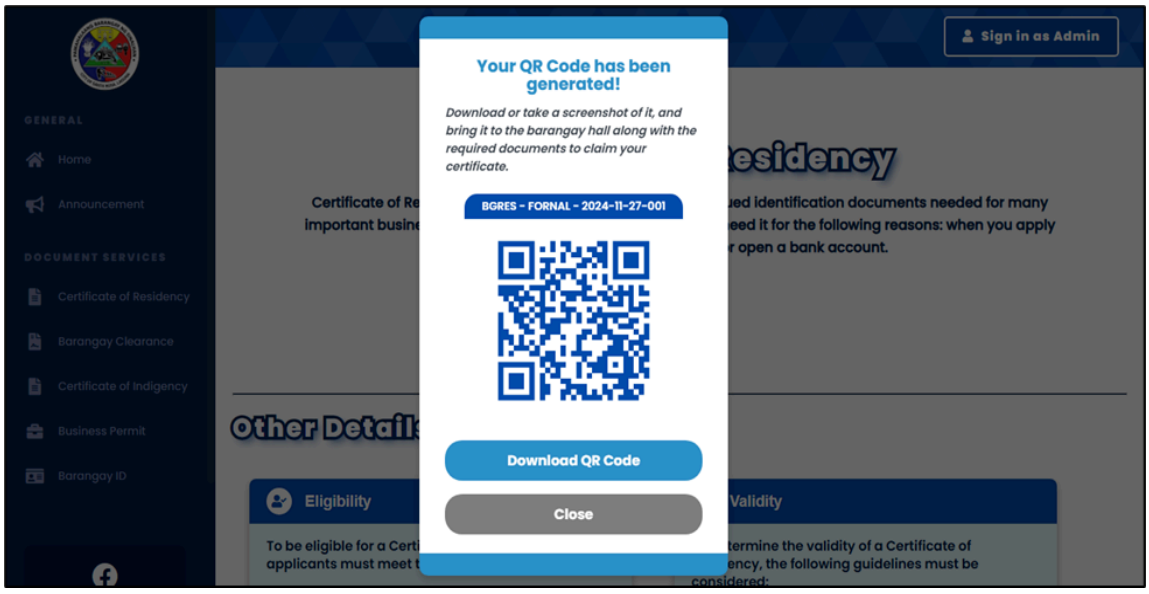
A "Request Form" button is located under the Other Details section. When clicked, a popup form specific to the chosen document type appears. The form contains fields for residents to complete with their personal details, which are customized based on the document type to align with the required information for generating the document. Each input field includes a validation check to ensure the accuracy of the information entered, and all fields are marked as required to prevent any critical information from being overlooked before submission.

Figure 7. Terms and Conditions Popup



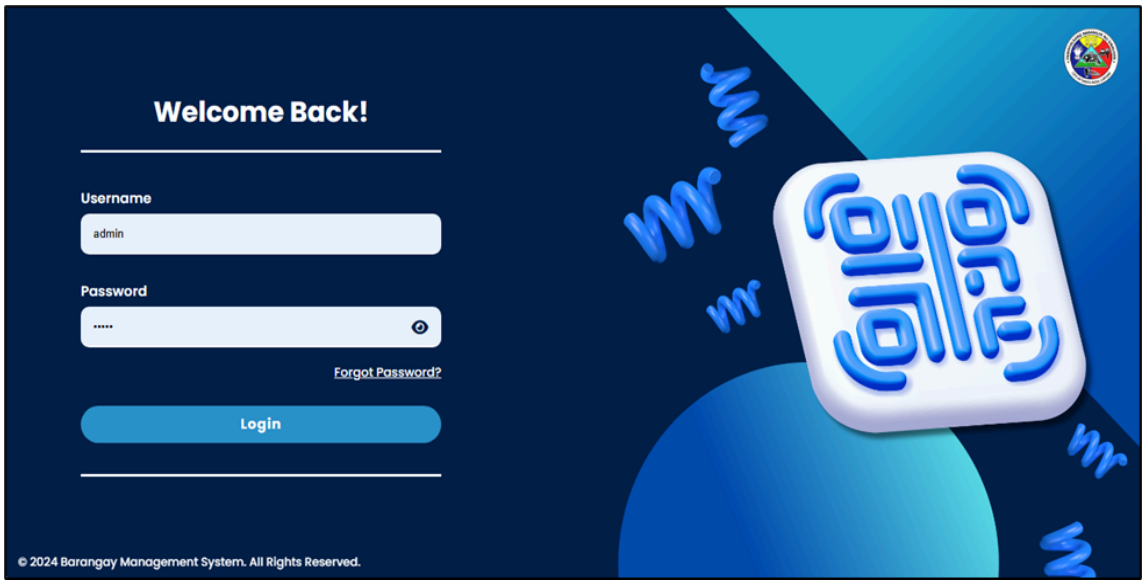
Before submitting the request form, residents are required to read and accept the Terms and Conditions and Privacy Policy. These documents outline important information about the rules, responsibilities, and rights related to requesting the document, as well as how personal information will be collected, used, and protected. A checkbox is provided for the resident to acknowledge that they have read and understood these policies. When the Terms and Conditions or Privacy Policy are clicked, they will open and be displayed on the screen for review. Only after the resident checks the box to agree to these terms and the privacy policy will the form be allowed to be submitted, ensuring that the resident is fully informed before proceeding with the request.

Figure 8. Successful QR Code Generation Alert with Generated QR Code



Upon successful submission of the completed request form, a QR code is generated and displayed on the screen. This QR code contains the template of the requested document, pre-filled with the information provided by the resident in the form. A "Download" button is available, allowing the resident to download the QR code. The downloaded QR code, along with the necessary documents, must be presented at the barangay hall to receive the requested document.

Figure 9. Admin/Staff Login Page



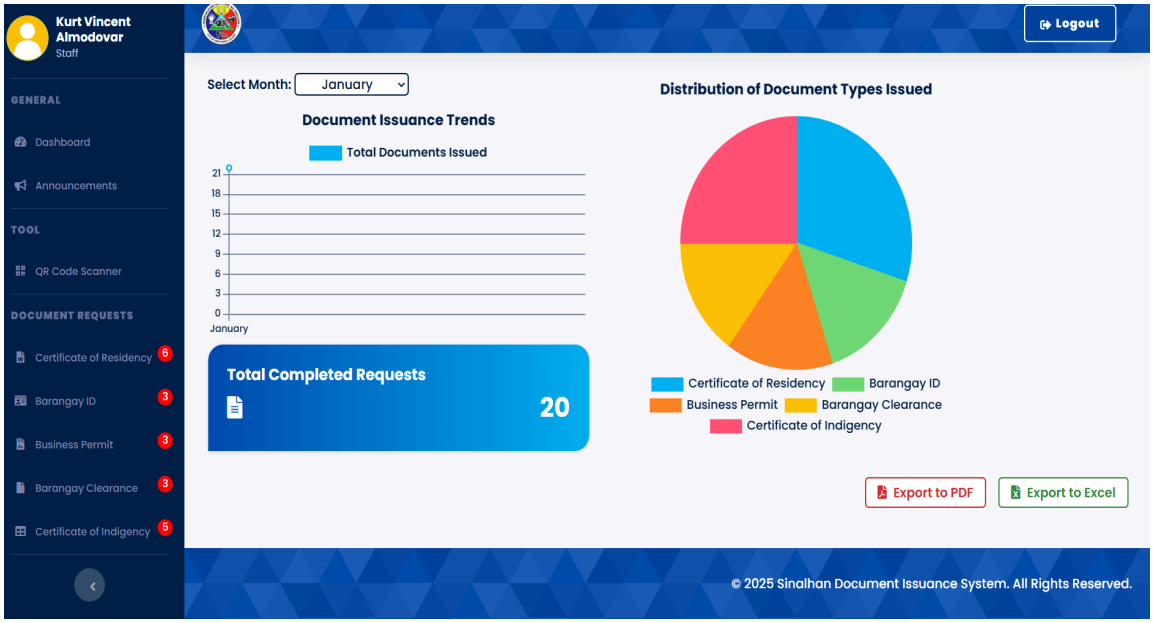
Once the "Sign in as Admin" button is clicked, the user will be redirected to the Login Page designed for both admin and staff members. On this page, users are required to enter their registered username and password. If the password is forgotten, a "Forgot Password?" link is provided, prompting the user to enter their registered email and username. A reset code will be sent to their email, allowing them to reset their password. After entering the correct credentials, users can click the “Login” button to access the system.

Figure 10. Admin Dashboard



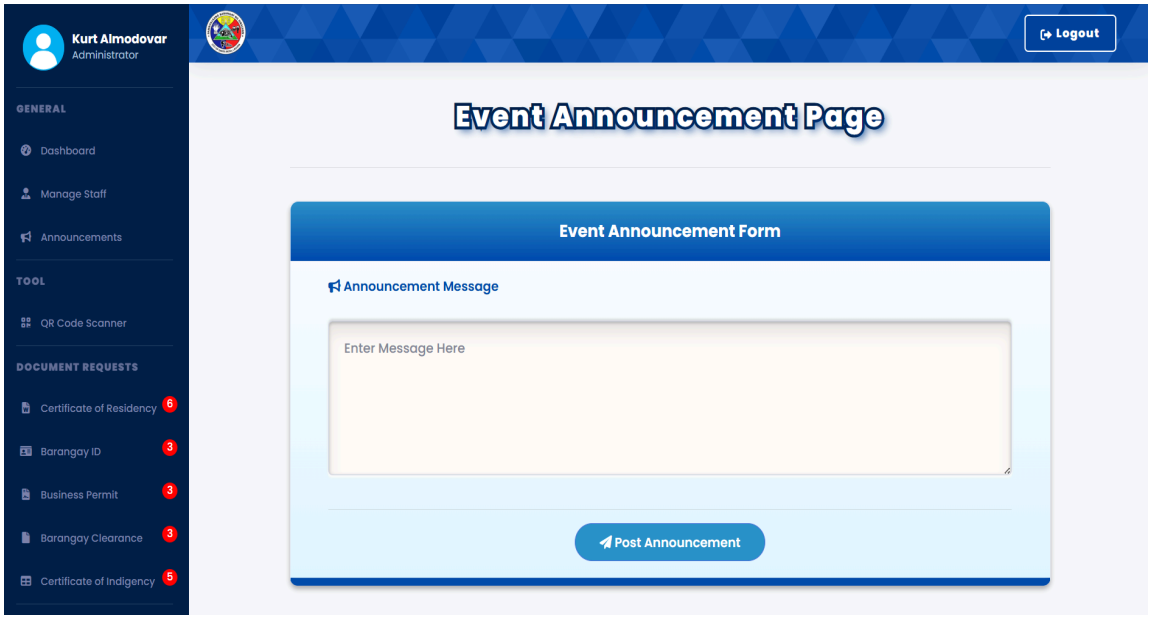
After a successful login, the user is redirected to the Dashboard. Both administrators and staff can access the export feature for generating daily reports, ensuring that essential records and activities are documented efficiently. The Dashboard includes a header with a “Logout” button and a fixed sidebar containing navigation links, applied across all pages. For administrators, the sidebar features a blue color-coded profile section displaying their name and role. The administrator dashboard also includes comprehensive statistics, such as the total count of staff, the total count of male staff, and the total count of female staff, displayed directly on the dashboard. Administrators can access management features, including the "Manage Staff" option, to oversee staff profiles. Notification numbers are displayed beside each document type, indicating the count of active requests that have not yet been archived.

Figure 11. Staff Dashboard



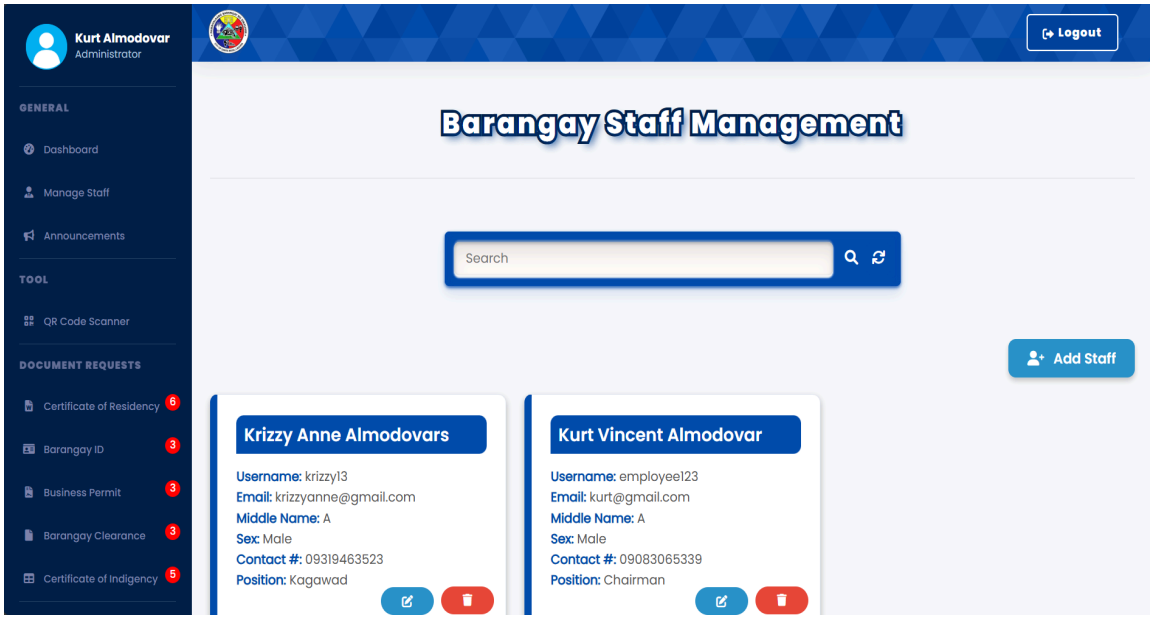
In contrast, the Staff Dashboard shares a similar layout with the header and sidebar, but the profile section is color-coded yellow. Staff members do not have access to the "Manage Staff" option and cannot manage other staff profiles. Additionally, the staff dashboard does not display staff statistics, such as the total number of staff or gender breakdown. However, staff members can view notification numbers for active requests beside each document type and also have the option to export data for daily reports.

Figure 12. Event Announcement Page



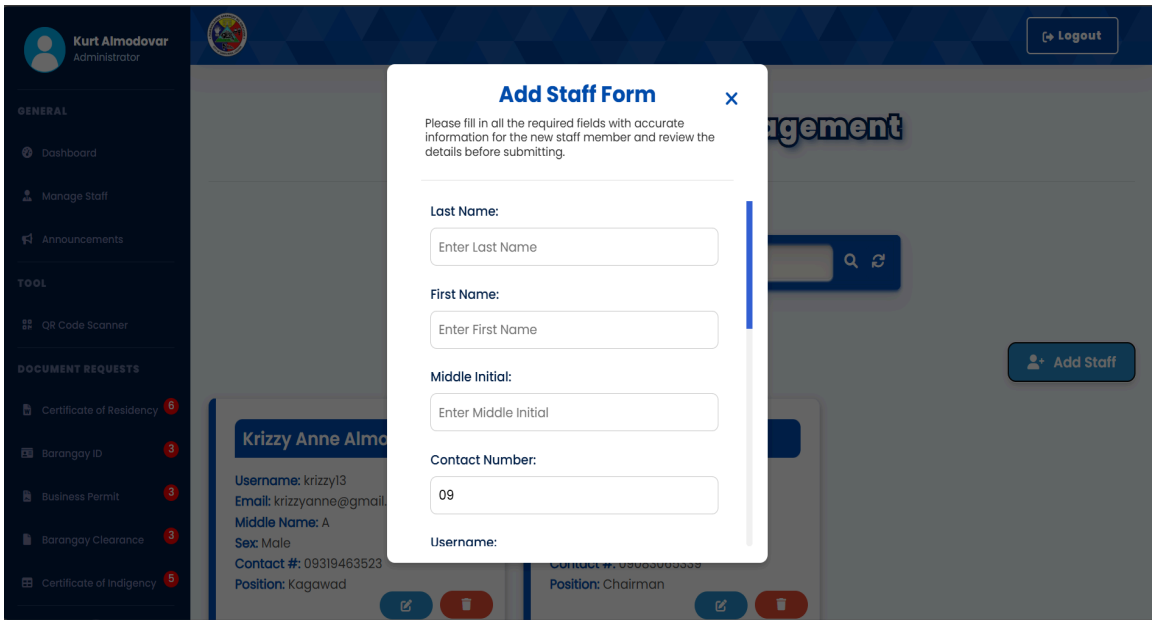
Both administrators and staff members can create and manage announcements through the Event Announcement Page. This page allows them to view all current announcements, including details such as who posted them and the date they were posted. They also have the option to remove announcements when necessary. Any announcement created on this page will automatically appear in the Announcement Section on the resident side, ensuring residents stay informed about the latest updates and events.

Figure 13. Barangay Staff Management Page



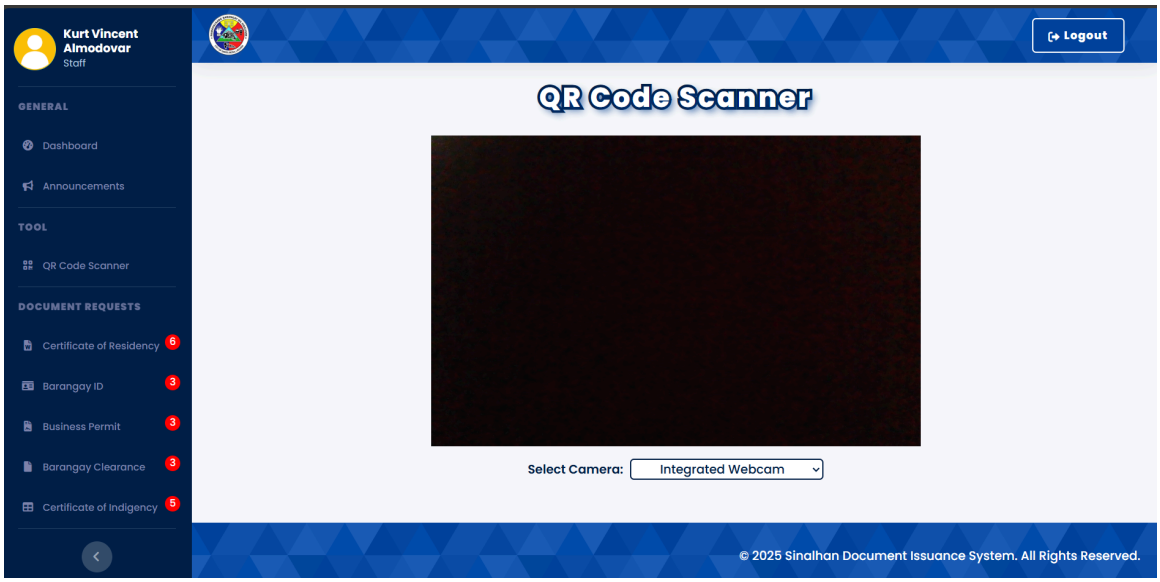
The Barangay Staff Management Page allows the administrator to easily manage staff members in the system. Administrators can update staff information, such as their details and roles, to keep records accurate. They can also remove staff who are no longer part of the barangay team. The page also lets administrators add new staff members by entering their information, making onboarding simple. This centralized tool helps ensure the barangay staff is well-organized and efficient.

Figure 14. Add Staff Form Popup



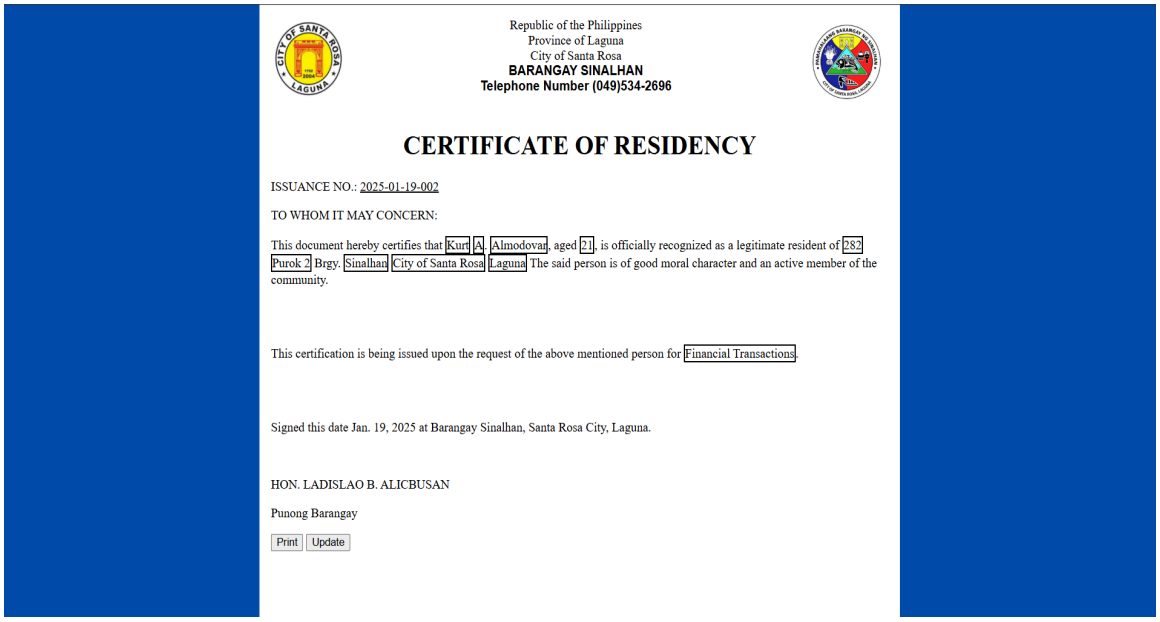
When the "Add Staff" button is clicked, a form will pop up containing fields related to the staff details that need to be added. This form includes validation checks to ensure the accuracy of the input, and all fields are marked as required to prevent any missing information. Once all fields are filled out completely, the administrator can click the "Submit" button, and the new staff member will be automatically registered in the system. A default password will also be assigned to the staff member upon successful registration.

Figure 15. QR Code Scanner Page



The system includes a QR Code Scanner Page, which can be accessed by both administrators and staff members to scan residents' QR codes. Users can connect a camera or a QR code scanner to their device and simply select the desired camera or scanner from the available options in the system.

Figure 16: Certificate of Residency with Supplied Information from QR Code Scan



Republic of the Philippines
Province of Laguna
City of Santa Rosa
BARANGAY SINALHAN
Telephone Number (049)534-2696

CERTIFICATE OF RESIDENCY

ISSUANCE NO.: 2025-01-19-002

TO WHOM IT MAY CONCERN:

This document hereby certifies that Kurt A. Almodovar, aged 21, is officially recognized as a legitimate resident of 282 Purok 2 Brgy. Sinalhan, City of Santa Rosa, Laguna. The said person is of good moral character and an active member of the community.

This certification is being issued upon the request of the above mentioned person for Financial Transactions.

Signed this date Jan. 19, 2025 at Barangay Sinalhan, Santa Rosa City, Laguna.

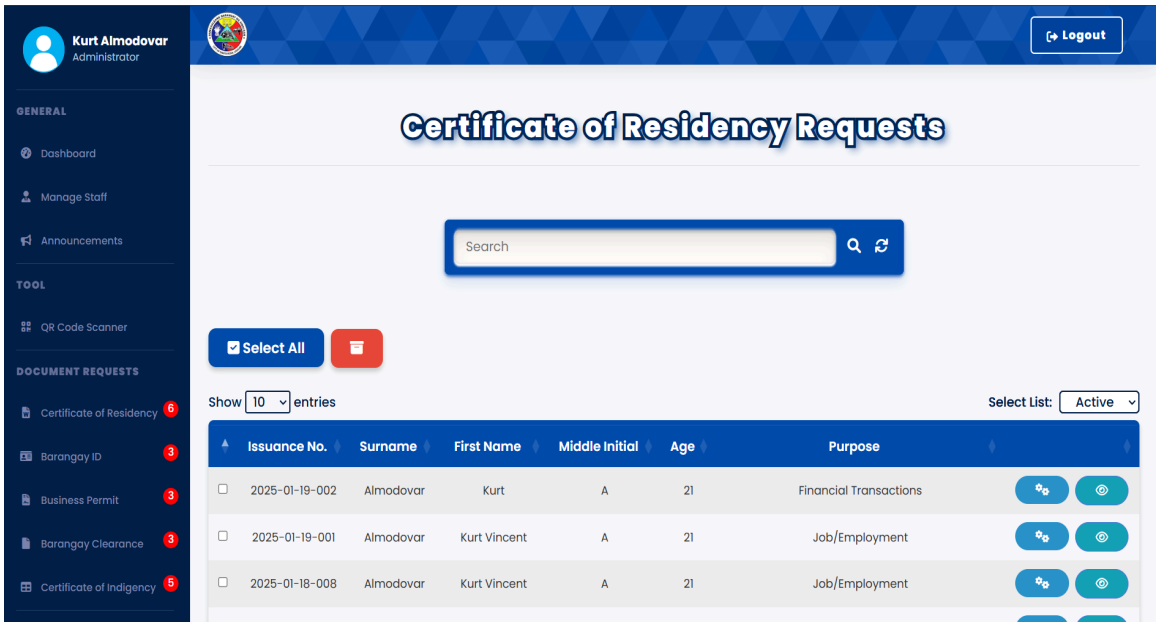
HON. LADISLAO B. ALICBUSAN

Punong Barangay

Print Update

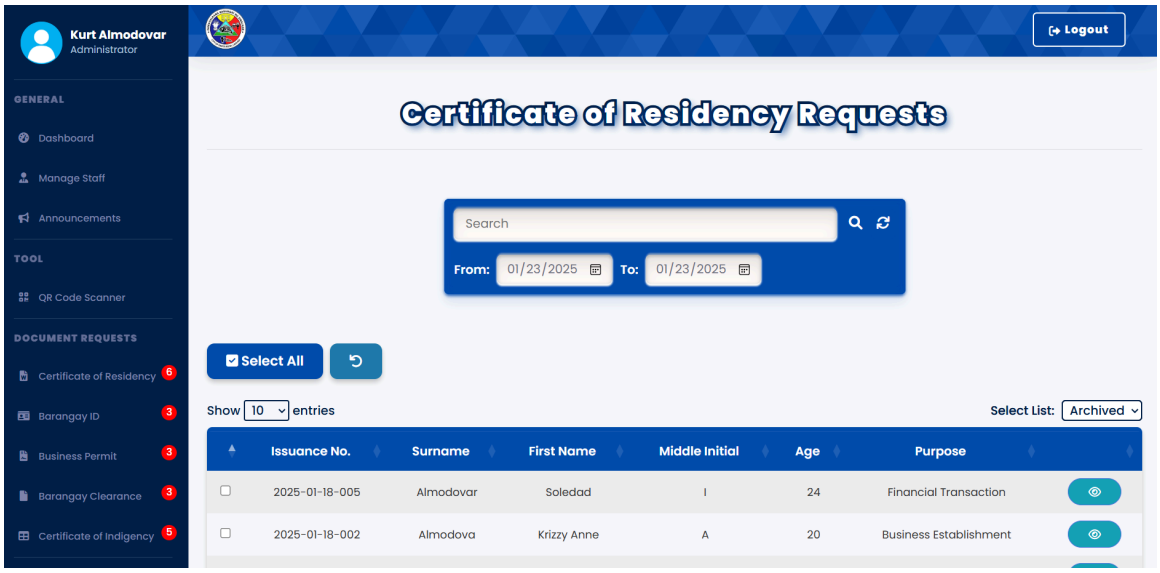
Once the QR code is scanned, it will redirect to the actual document request template, displaying the supplied information from the QR code. The specific type of document displayed depends on the document request chosen by the resident. This information is subject to verification to ensure its accuracy before the document request can be processed further."

Figure 17. Certificate of Residency Request List Page



Each type of document has its own webpage displaying a list of active requests. The number of active requests for each document type is shown on the notification beside the document name. Once a document request is fully processed or issued, both the administrator and staff members can archive it. Furthermore, both have the ability to search for specific resident information related to document requests. The search results will display a list of the resident's request details, allowing for quick access to relevant records.

Figure 18. Archived Request List



The archived requests are moved to an Archive List and can be retrieved whenever needed. The archived list can be exported as either a PDF or Excel file for record-keeping and further processing. Additionally, both the admin and staff members have the ability to search for specific resident information related to their document requests. The search results will display a list of the resident's request details, allowing for quick access to relevant records.